

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **6224**

AY:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C Code:	CS413
Exam Type:	Written	Block:	202	Seat No:	2022600005
		Course Section:	1		

L_61_463_986_L_1505_48208



Name of the candidate. **BYANDSHALI HARSHIT RAJESH**

Candidate's UID number: **2022600005**

Examination class room number: **202**

Invigilator's signature with date: **08/10/2025**

(To be filled in by the candidate)

EXAMINATION: **B.E. BTECH** **MSE**
(For example FY MCA/FY BTECH)

SEMESTER : **VII**

PROGRAM: **CSE (AIML)**

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: **WEDNESDAY, 8/10/2025**

COURSE NAME: **DEEP LEARNING**

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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No.

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(Begin answer for each question on a new page)

1 a)

$$f(w) = \frac{1}{2} |xw - y|^2$$

$$\cancel{f} w = w_{old} - \eta \frac{\partial f(w)}{\partial w}$$

$$\therefore w_{new} = w_{old} - \eta \frac{\partial \left(\frac{1}{2} |xw - y|^2 \right)}{\partial w}$$

$$= w_{old} - \frac{\eta}{2} (2 (xw - y))$$

$$w_{new} = w_{old} - \frac{\eta}{2} (2 (xw - y))$$

$$\cancel{\frac{\partial f}{\partial w}} w_{new}$$

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Question No.	Answer
1	1. The first step in the process of creating a new product is to identify a market need. This is often done through market research, which can involve surveys, focus groups, and other methods of gathering information about consumer preferences and behaviors.
2	2. Once a market need has been identified, the next step is to develop a concept for the new product. This involves brainstorming ideas and determining the key features and benefits of the product.
3	3. The third step is to create a prototype of the product. This can be done using a variety of methods, including 3D printing, computer-aided design (CAD), and traditional manufacturing techniques.
4	4. The fourth step is to conduct a feasibility study. This involves evaluating the technical, financial, and market viability of the product concept.
5	5. The fifth step is to develop a business plan for the new product. This includes determining the target market, pricing strategy, distribution channels, and other key business factors.
6	6. The sixth step is to secure funding for the product development process. This can be done through a variety of sources, including venture capital, angel investors, and crowdfunding.
7	7. The seventh step is to manufacture the product. This involves setting up a production line, sourcing materials, and hiring workers.
8	8. The eighth step is to distribute the product. This involves finding retail partners, establishing a distribution network, and promoting the product.
9	9. The ninth step is to monitor the product's performance in the market. This involves tracking sales, customer feedback, and other key metrics.
10	10. The tenth step is to iterate on the product based on feedback and market performance. This involves making improvements to the product design, features, and marketing strategy.

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Question
No.

1 b)

$$\text{Error} = \text{Bias}^2 + \text{Variance} + \text{Noise}.$$

i) When capacity of model is too high, it means the model is too complex.

ii) Which means it has a lot of parameters to learn. So instead of learning parameters it directly starts learning the data itself.

iii) This results in high variance and overfitting. So the model gets high training accuracy but low test accuracy and model fails to generalize the data.



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Question
No.

2 b)

i) In an ~~an~~ undercomplete autoencoder the dimensions of the hidden layer are lesser than the input layer dimensions.

ii) Now since the dimensions are lesser the layer needs to learn only the important features. This is because there are not enough neurons to learn each and every feature of the image. And if ~~the~~ it learns the unimportant features then the output ~~result~~ image won't be the one that is expected which would give high errors.

iii) So if MSE (mean square error) is considered then it will force the autoencoder to learn other features. So basically only important features would be learnt and dimensions of latent space vectors would be less. Hence, it works similar to PCA but has an advantage of being non-linear.

iv) In case of over complete autoencoders the dimensions of hidden layer are more than the input layer ~~neurons~~.

v) This could lead to autoencoder learning the whole image instead of generalizing.

vi) But if we regularize the autoencoder, then it will not lead to learning the identity function.

vii) L_1 regularization would make the unimportant weights to 0 and L_2 would bring the ~~the~~ unimportant weights close to 0, avoiding the autoencoder to learn the identity function.

viii) Other than that ~~in~~ by adding sparsity we can turn off some neurons forcing other neurons to learn the important features, we could add noise to input to make autoencoder predict the correct output and make constructive autoencoder to make it more stable and resistant to changes.

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3 b)

GRU equations

$$u_t = \sigma(W_{uh} \times h_{t-1} + W_{ux} \times X + b_u) \quad \text{--- (1)}$$

$$r_t = \sigma(W_{rh} \times h_{t-1} + W_{rx} \times X + b_r) \quad \text{--- (2)}$$

$$\tilde{h}_t = \tanh(W_{hh} \times r_t \times h_{t-1} + W_{hx} \times X + b_h) \quad \text{--- (3)}$$

$$h_t = u_t \tilde{h}_t + (1 - u_t) h_{t-1} \quad \text{--- (4)}$$

~~and~~

Above are the equations of a GRU.

 ~~u_t~~ and ~~r_t~~ are the equations of update and reset gate.

② equation shows how candidate hidden state is dependent on reset gate and

④ equation shows how hidden state is dependent of previous hidden state, candidate hidden state and ~~the~~ the update gate.LSTM equations

$$f_t = \sigma(W_{fh} \times h_{t-1} + W_{fx} \times X + b_f) \quad \text{--- (5)}$$

$$i_t = \sigma(W_{ih} \times h_{t-1} + W_{ix} \times X + b_i) \quad \text{--- (6)}$$

$$o_t = \sigma(W_{oh} \times h_{t-1} + W_{ox} \times X + b_o) \quad \text{--- (7)}$$

$$\tilde{c}_t = \tanh(W_{ch} \times h_{t-1} + W_{cx} \times X + b_c) \quad \text{--- (8)}$$

$$c_t = f_t \times c_{t-1} + i_t \tilde{c}_t \quad \text{--- (9)}$$

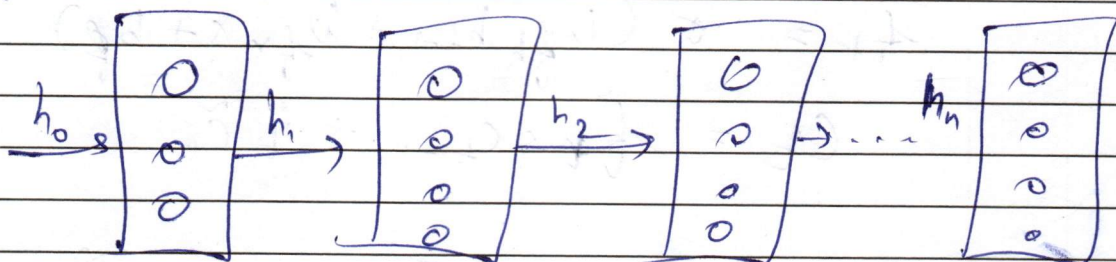
$$h_t = o_t \tanh(c_t)$$

Here as we can see in GRU by equations ①, ② and ③ there only 6 weight matrices (parameters) to be learnt but in LSTM by equations ⑤, ⑥, ⑦ & ⑧ there are 8 weight matrices (parameters) to be learnt.

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3 a



$$h_t = f(w_{hh} h_{t-1} + w_{hx} x + b_h) \quad \text{--- (1)}$$

$$y_t = g(w_{yh} x h_t + w_{yx} x + b_y) \quad \text{--- (2)}$$

$$L = - \sum y_i \log \hat{y}_i + (1 - y_i) \log (1 - \hat{y}_i)$$

$\therefore L$ is dependent on y .
By equations (1) and (2)

y is dependent on h_t .

And by equation (1) h_t is dependent on h_{t-1} .
 h_t is also dependent on w_{hh} .

$$\therefore L \rightarrow y \rightarrow h_t \rightarrow w_{hh}$$

$$\quad \quad \quad \downarrow$$

$$h_{t-1} \rightarrow w_{hh}$$

$$\quad \quad \quad \downarrow$$

$$h_{t-2} \rightarrow w_{hh}$$

This is how L is recursively dependent.

$$\therefore \frac{\partial L}{\partial h_t} = \frac{\partial L}{\partial y} \frac{\partial y}{\partial h_t} + \frac{\partial L}{\partial y} \frac{\partial y}{\partial h_t} \left(\frac{\partial h_t}{\partial w_{hh}} + \frac{\partial h_t}{\partial h_{t-1}} \right)$$

$$+ \frac{\partial L}{\partial y} \frac{\partial y}{\partial h_t} \frac{\partial h_t}{\partial h_{t-1}} \frac{\partial h_{t-1}}{\partial w_{hh}} + \dots$$

$$= \frac{\partial L}{\partial y} \frac{\partial y}{\partial h_t} \left(\frac{\partial h_t}{\partial w_{hh}} + \frac{\partial h_t}{\partial h_{t-1}} \frac{\partial h_{t-1}}{\partial w_{hh}} + \dots \right)$$

$$\frac{\partial L}{\partial h_{t-k}} = \frac{\partial L}{\partial y} \frac{\partial y}{\partial h_t} \prod_{k=1}^t \frac{\partial h_t}{\partial h_{t-k}}$$

This gradient can grow too small resulting in vanishing gradient problem.

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3

a)

$$f_t = \sigma(w_{fh} h_{t-1} + w_{fx} x_t + b_f)$$

$$C_t = f_t \times C_{t-1} + i_t \tilde{C}_t$$

In LSTM two states are maintained, cell state and hidden state.

And forget gate controls if we want to forget or propagate C_t .

Here instead of multiplication C_t is dependent on additions which does not reduce the gradient so much hence vanishing gradient is prevented.

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